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(54) **PHOTOPLETHYSMOGRAPHY SENSORS
AND PROCESSES**

(71) Applicant: **Huxley Medical, Inc.**, Atlanta, GA
(US)

(72) Inventors: **Brennan Torstrick**, Atlanta, GA (US);
Mohsen Safaei Mohammadabadi,
Smyrna, GA (US)

(73) Assignee: **HUXLEY MEDICAL, INC.**, Atlanta,
GA (US)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

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Aug. 24, 2022, now Pat. No. 12,279,853.

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A61B 5/00 (2006.01)
A61B 5/024 (2006.01)
A61B 5/282 (2021.01)

(52) **U.S. Cl.**
CPC **A61B 5/02416** (2013.01); **A61B 5/282**
(2021.01); **A61B 5/6833** (2013.01); **A61B**
2562/164 (2013.01); **A61B 2562/166** (2013.01)

(58) **Field of Classification Search**

CPC A61B 5/1455
See application file for complete search history.

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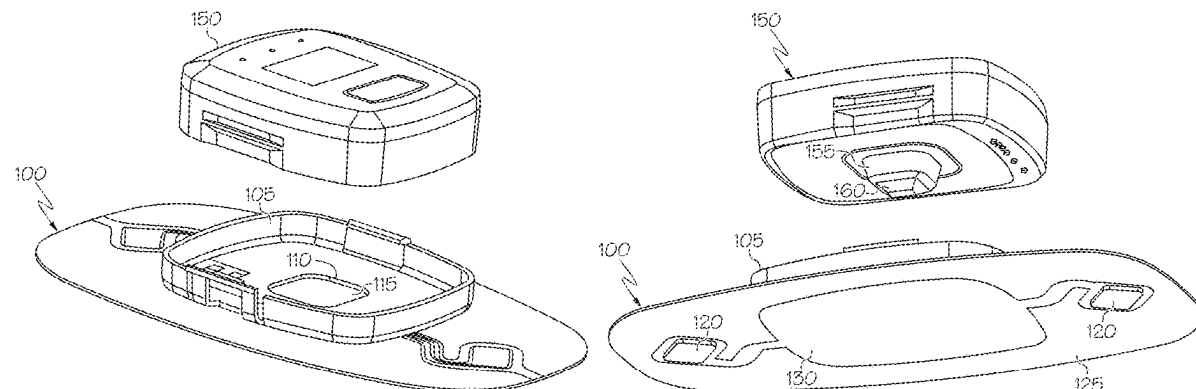
Primary Examiner — Marjan Fardanesh

(74) *Attorney, Agent, or Firm* — Husch Blackwell LLP;
Bryan D. Stewart

(57) **ABSTRACT**

The present disclosure relates to devices and processes that produce superior photoplethysmography (PPG) signals. In various embodiments, the present devices and processes leverage load backing, conformal contact, and strain isolation mechanisms to produce higher quality and amplitude PPG signals, and provide more repeatable results than previous devices and processes. In at least one embodiment, the devices discussed herein include a rigid support structure and elastic spacer for supporting a PPG sensor and various adhesive layers.

20 Claims, 15 Drawing Sheets



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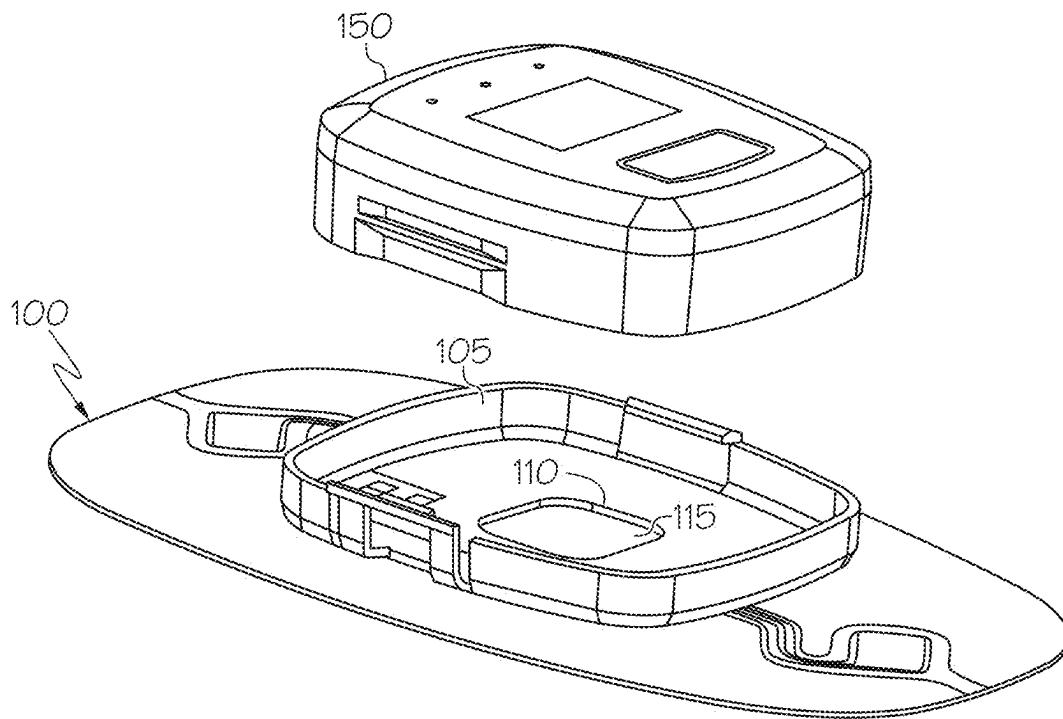


FIG. 1A

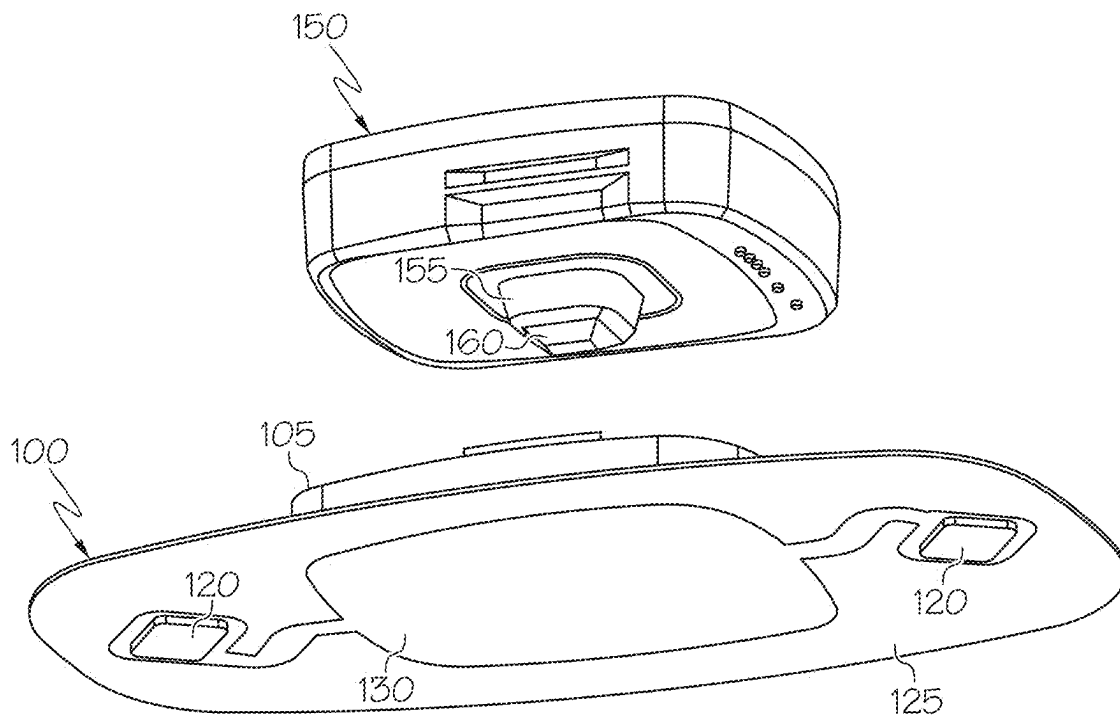


FIG. 1B

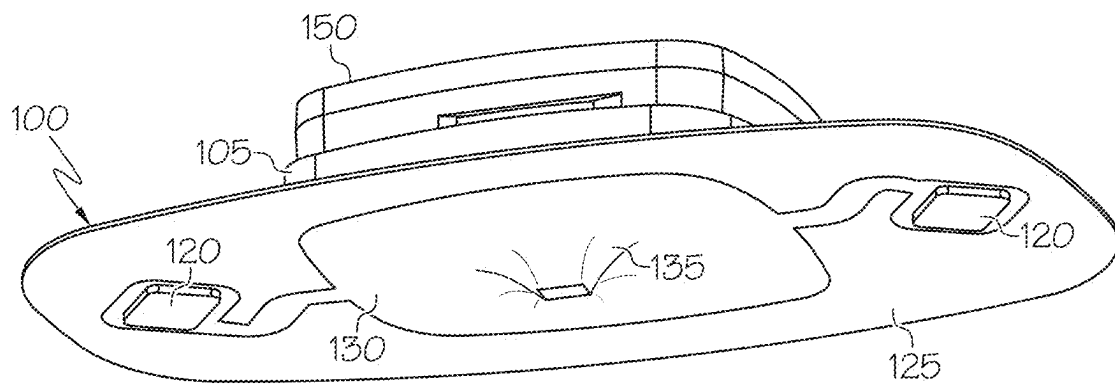


FIG. 1C

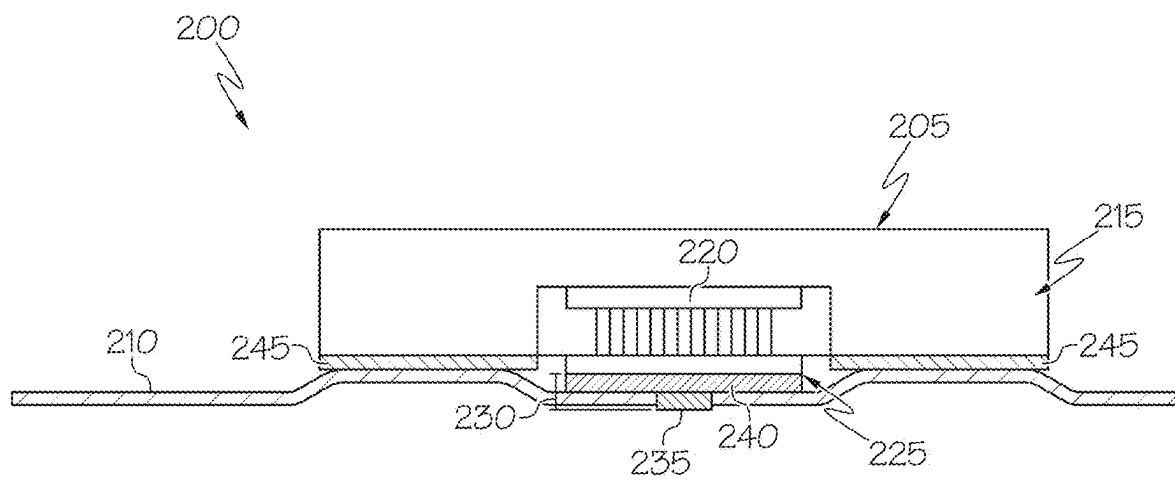


FIG. 2

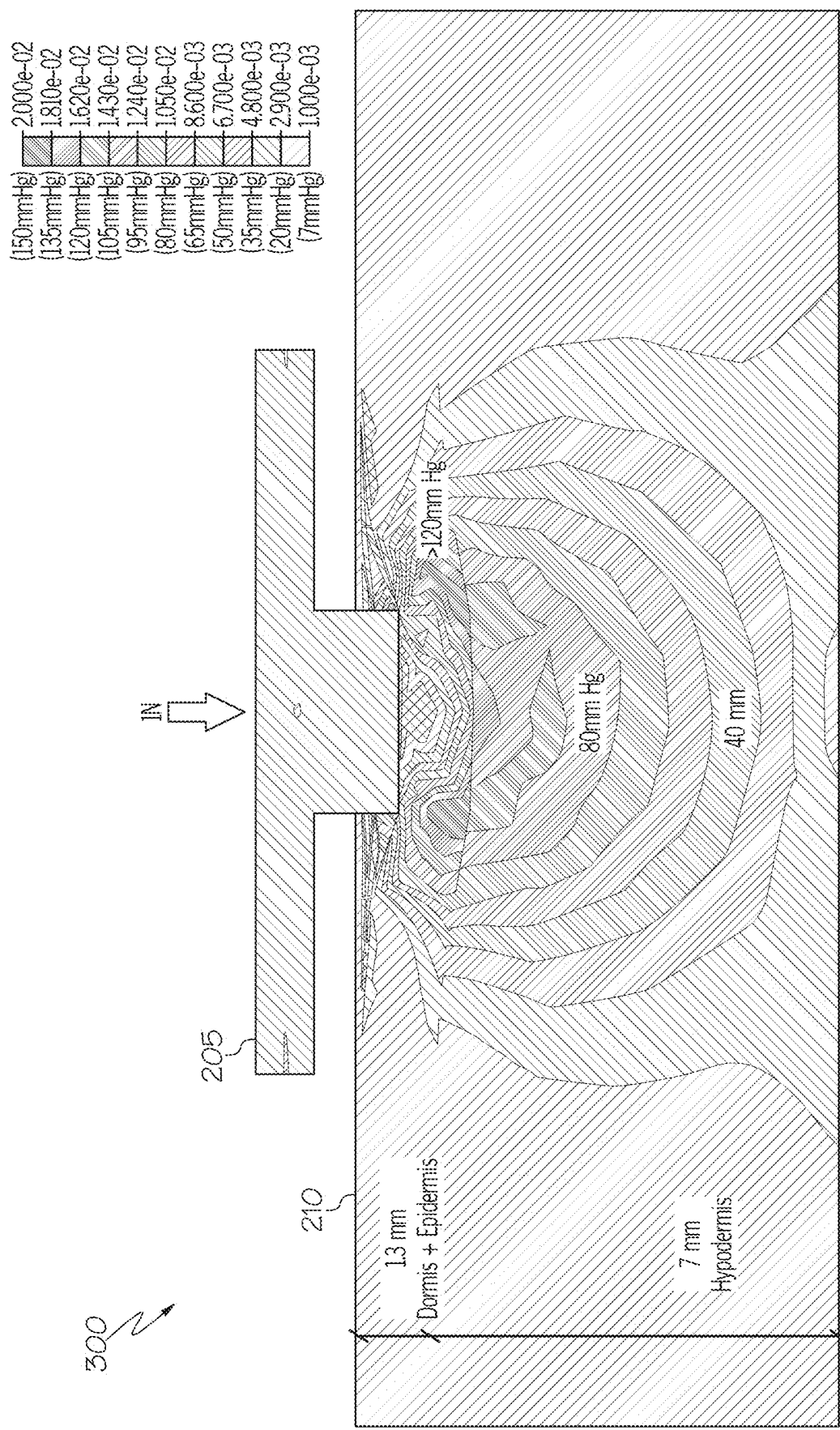


FIG. 3

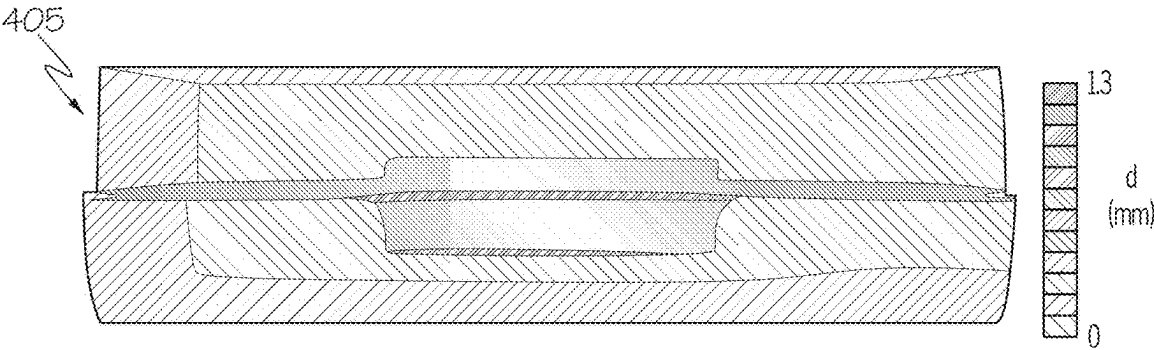


FIG. 4A

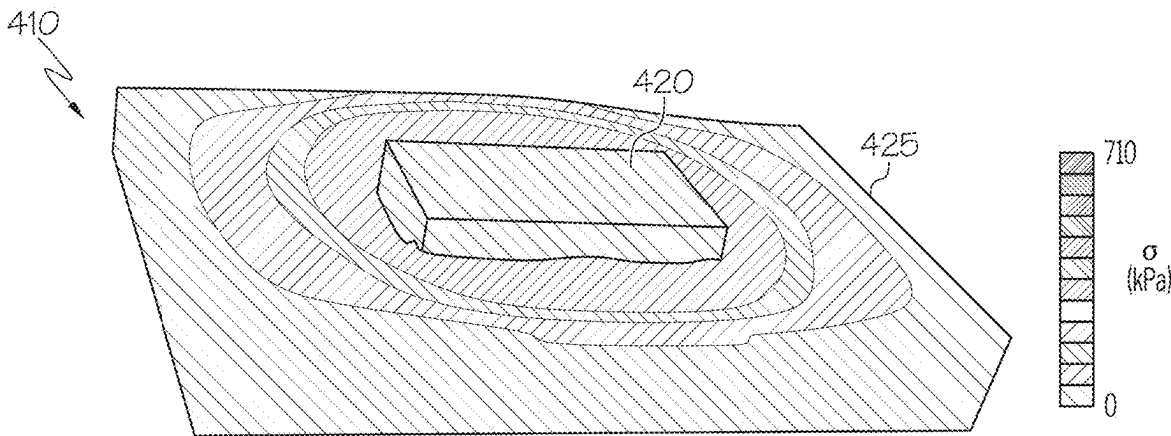


FIG. 4B

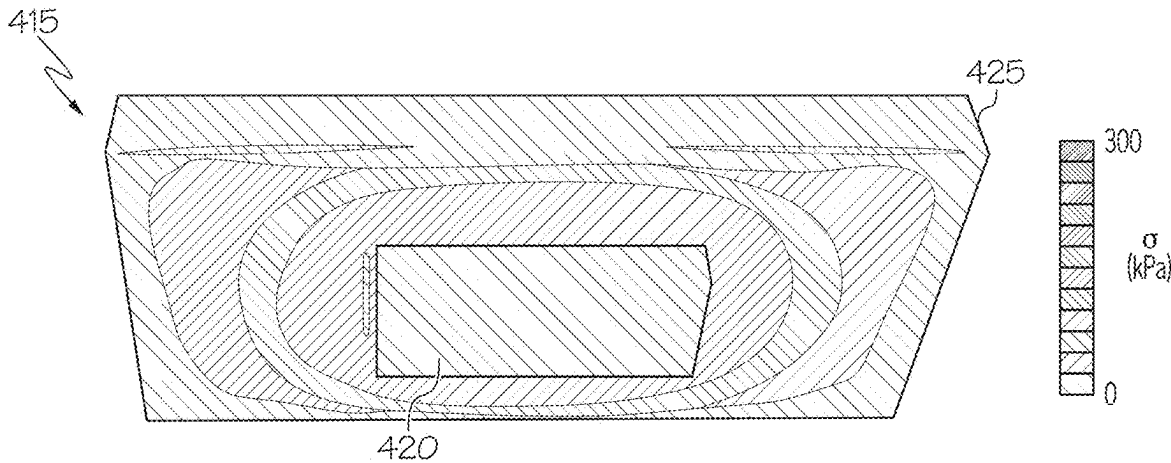


FIG. 4C

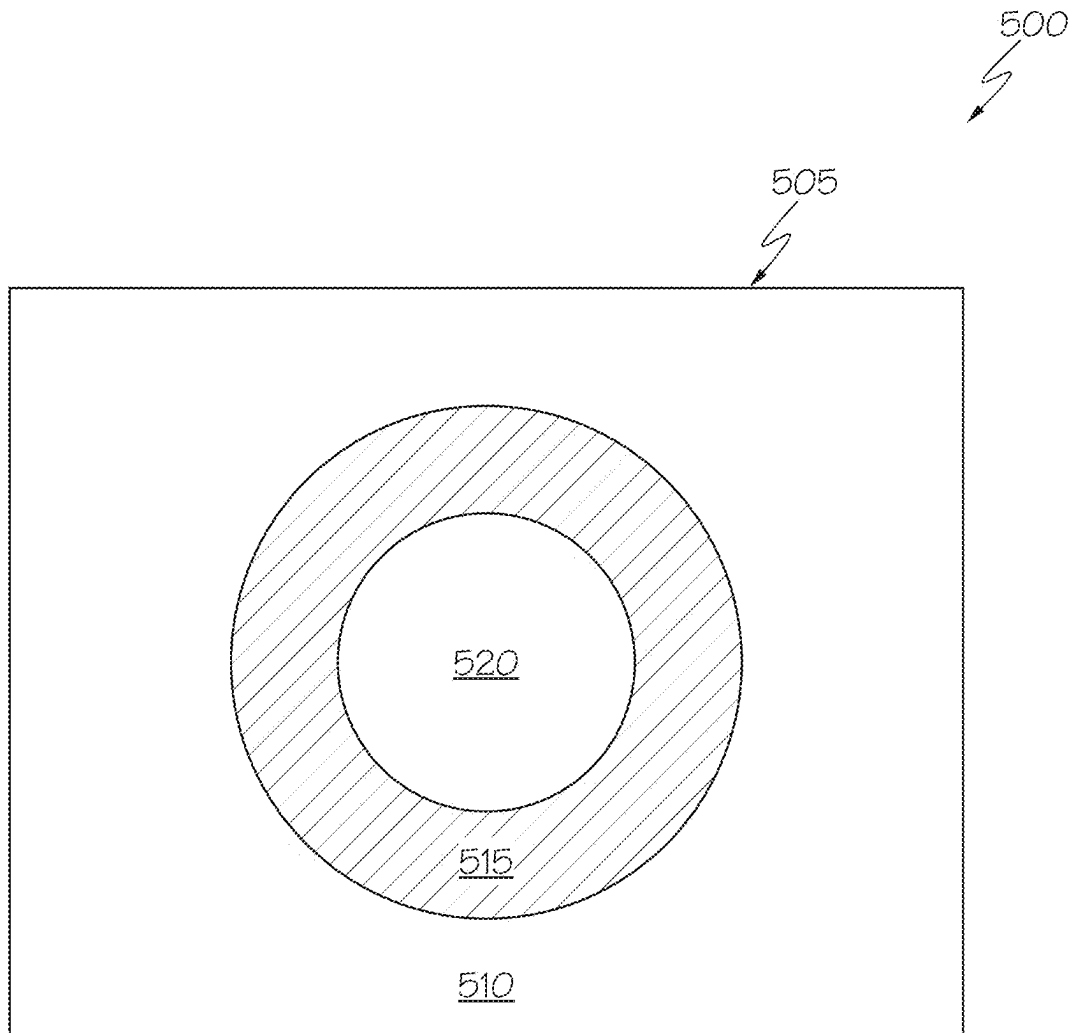


FIG. 5

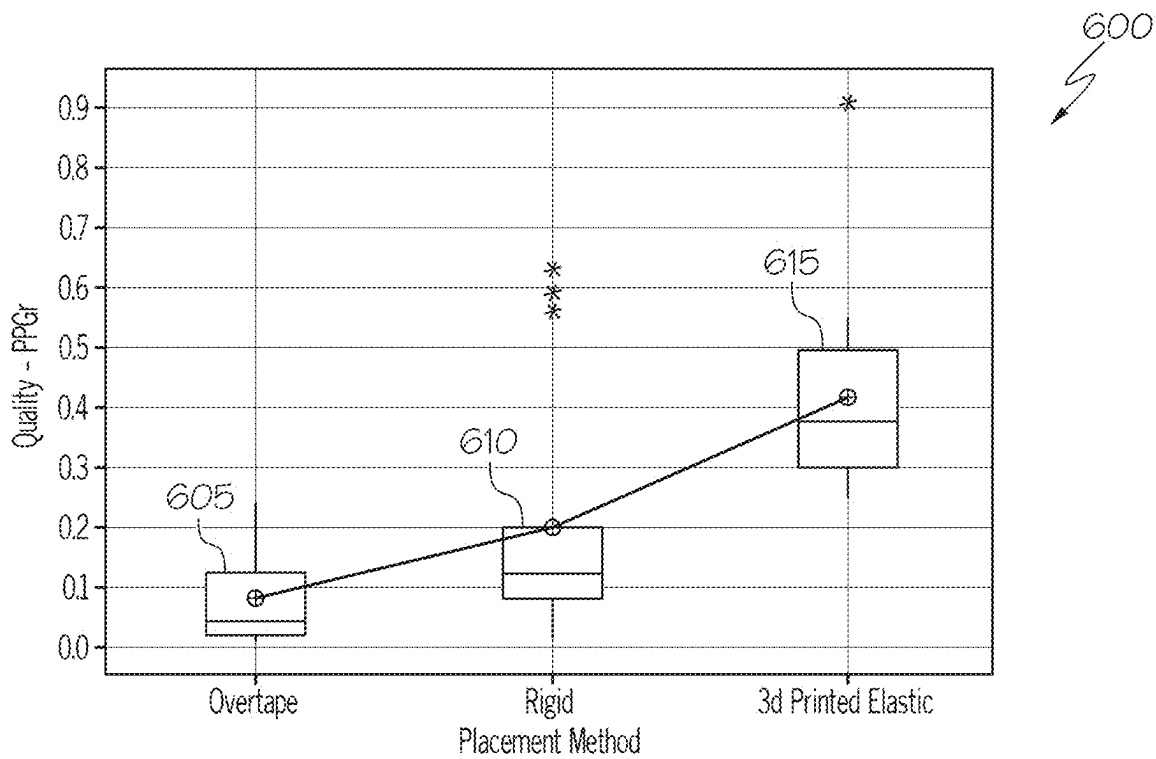


FIG. 6A

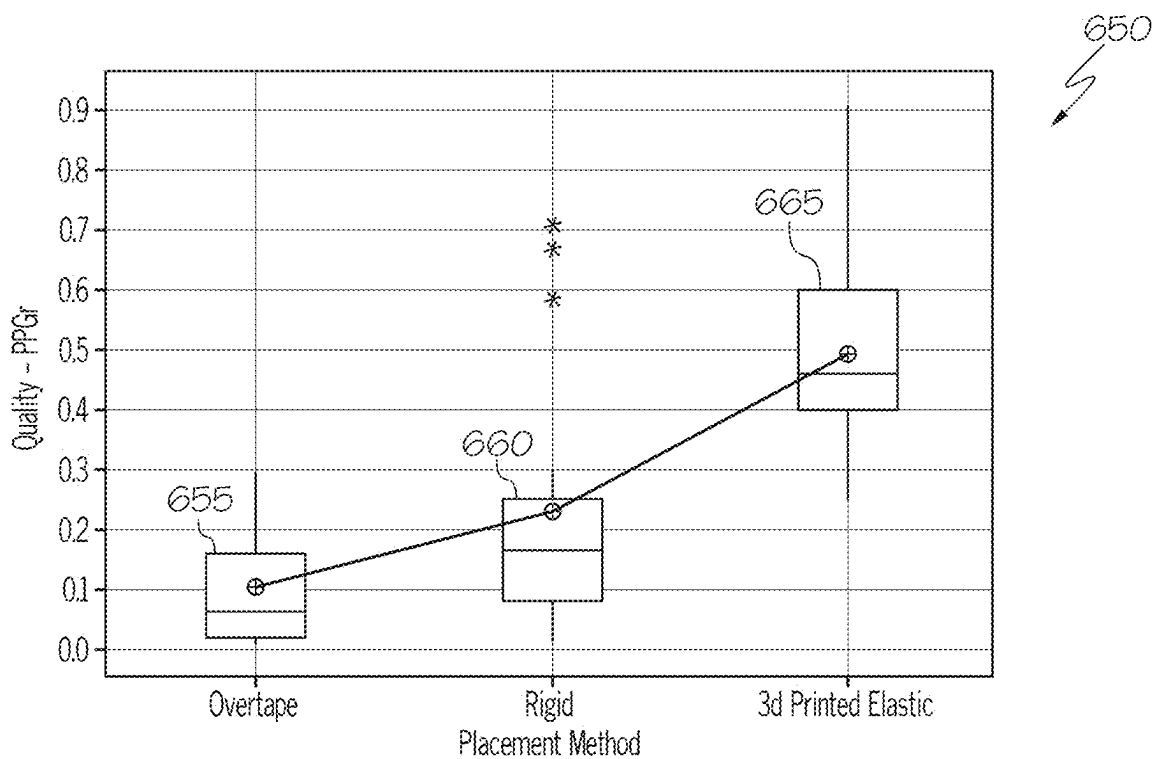


FIG. 6B

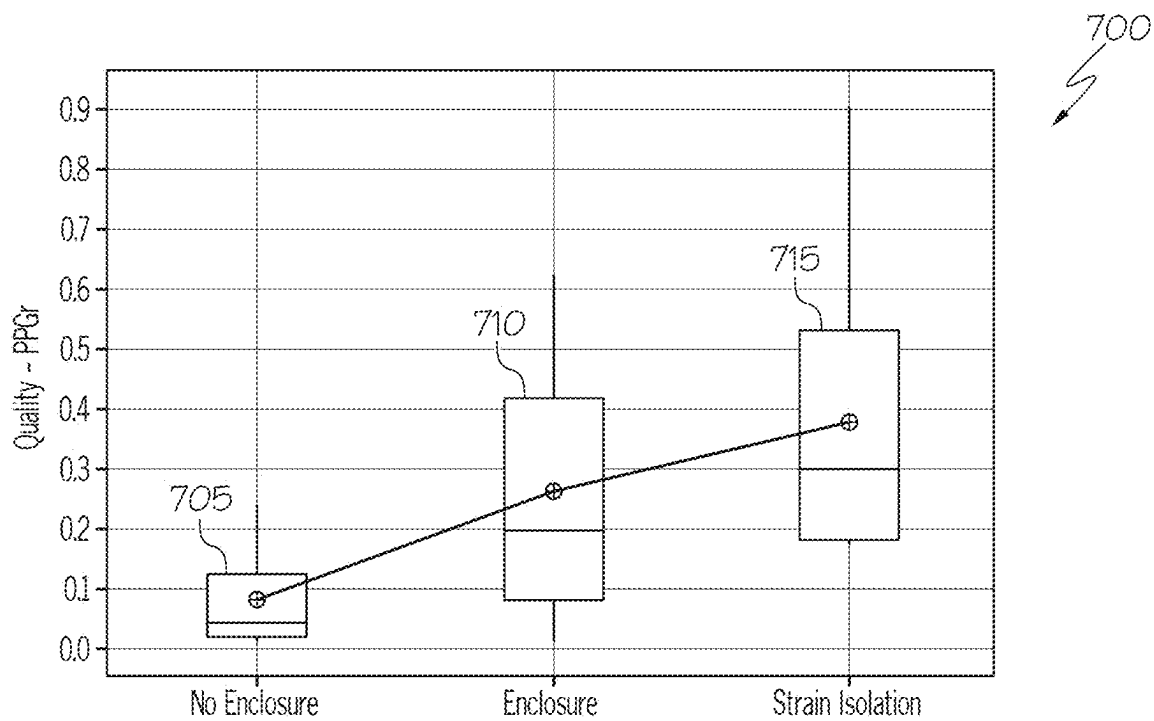


FIG. 7A

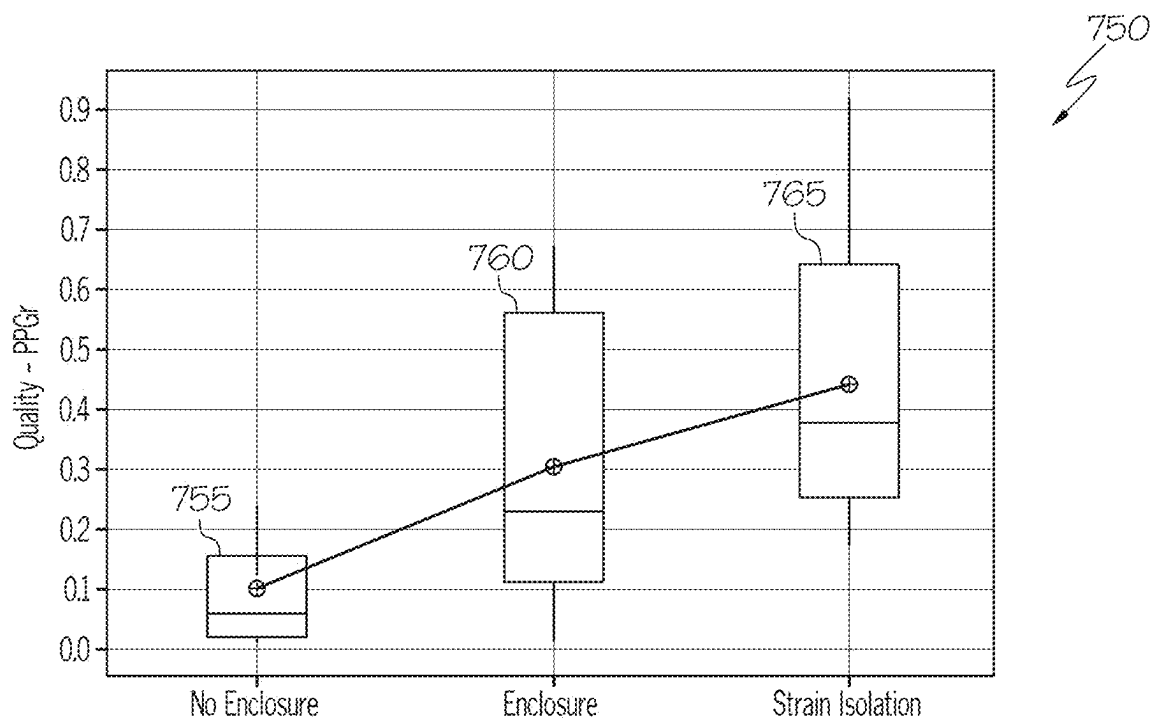


FIG. 7B

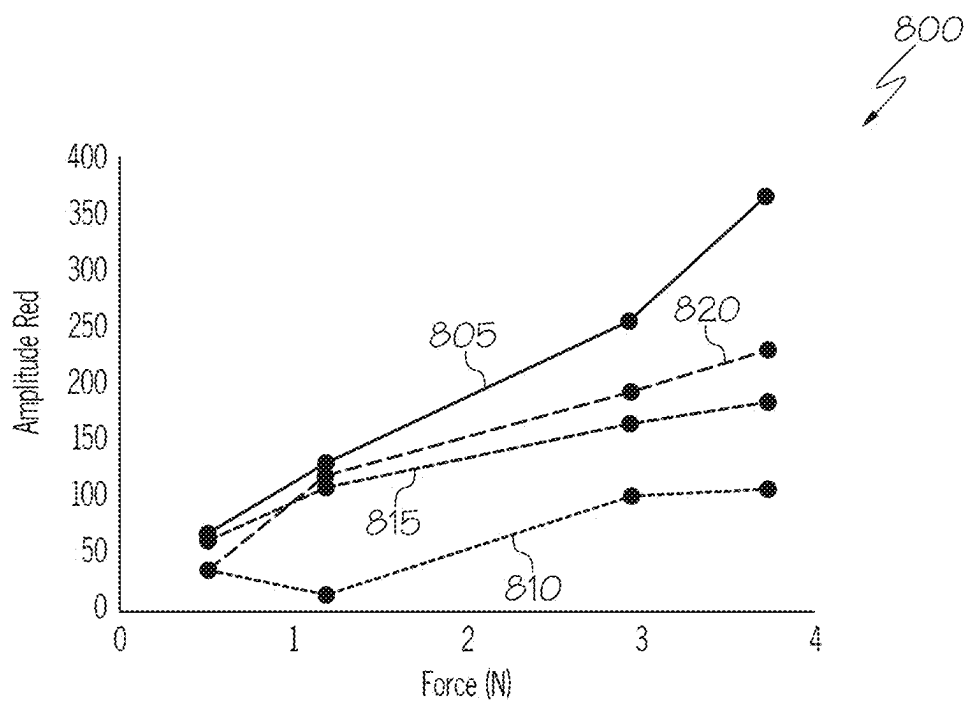


FIG. 8A

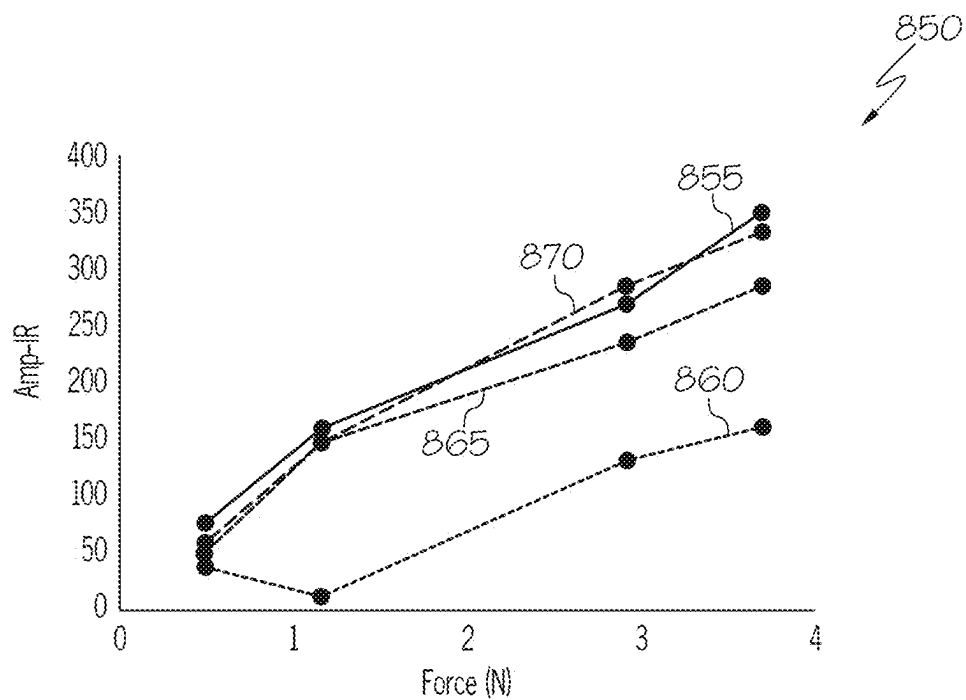


FIG. 8B

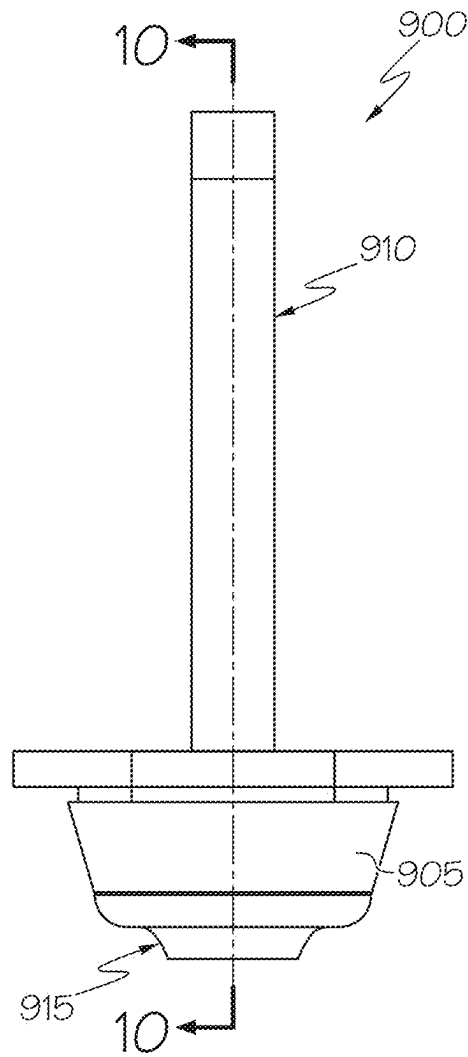


FIG. 9

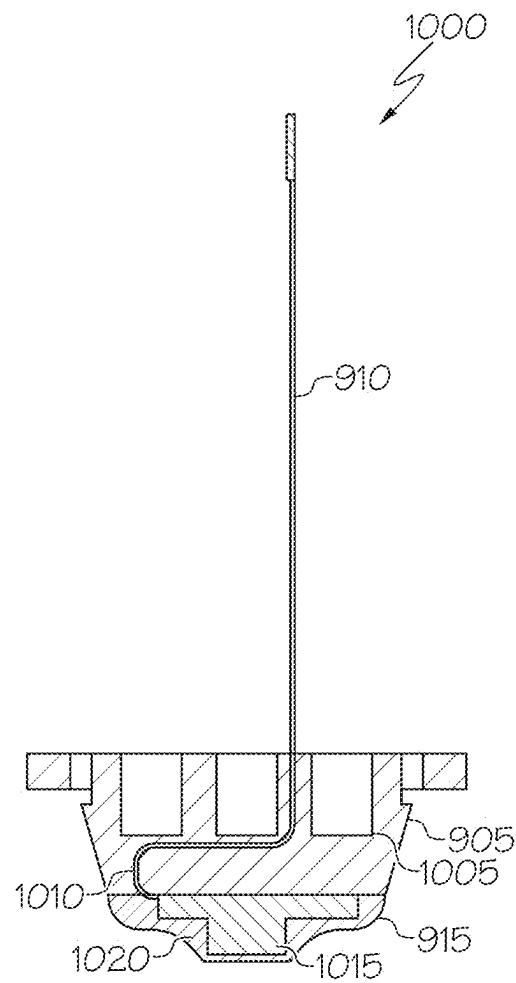


FIG. 10

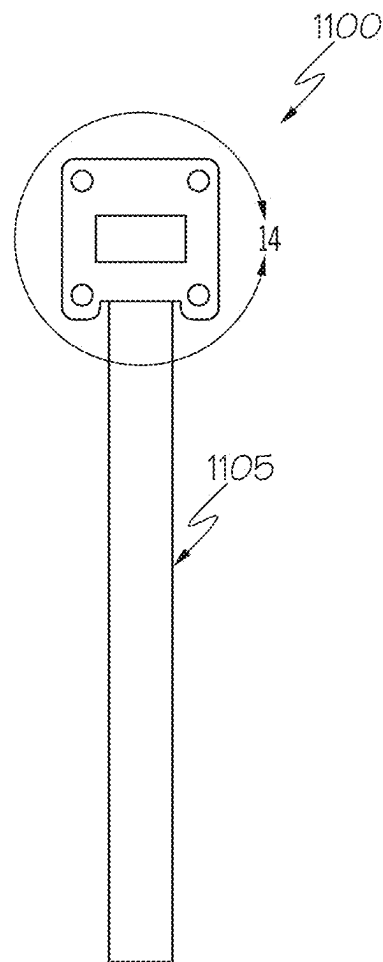


FIG. 11

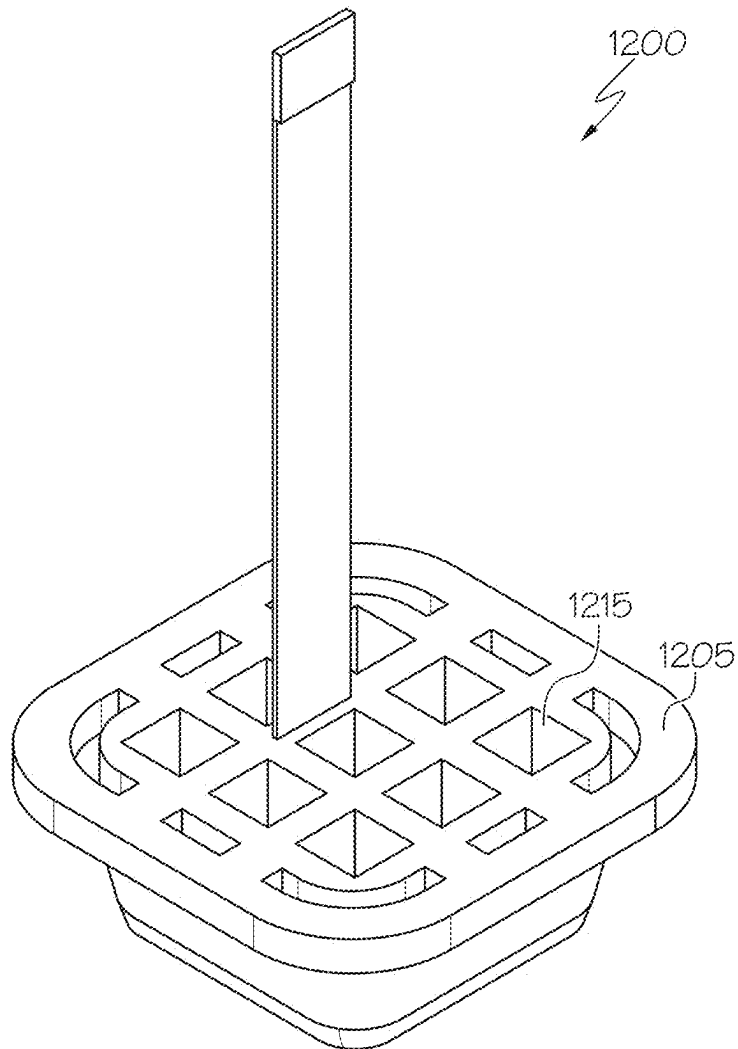


FIG. 12

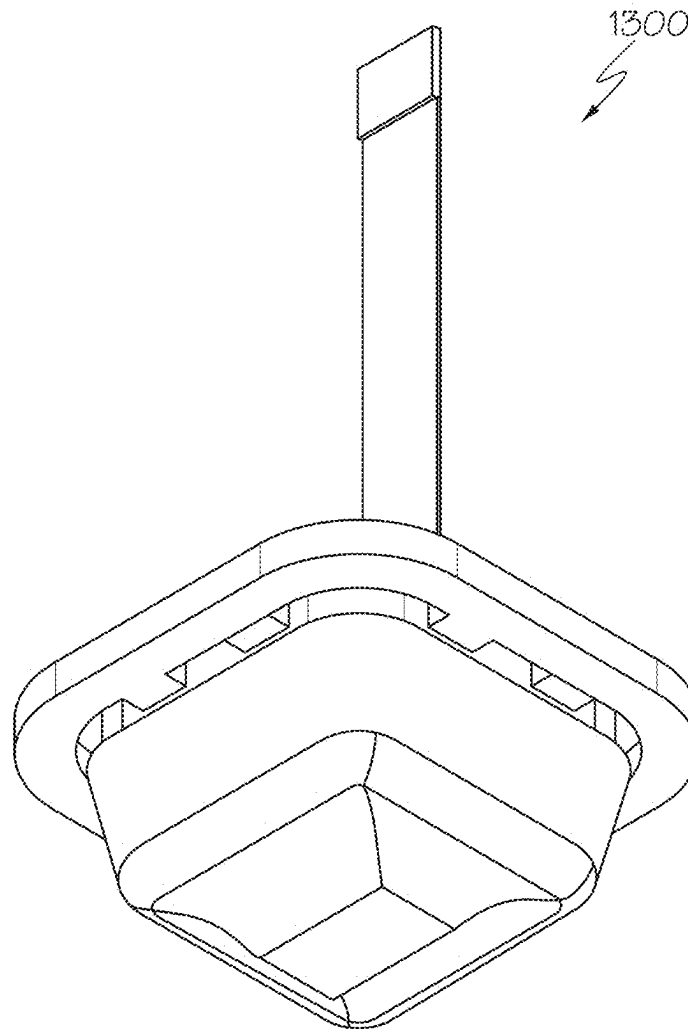


FIG. 13

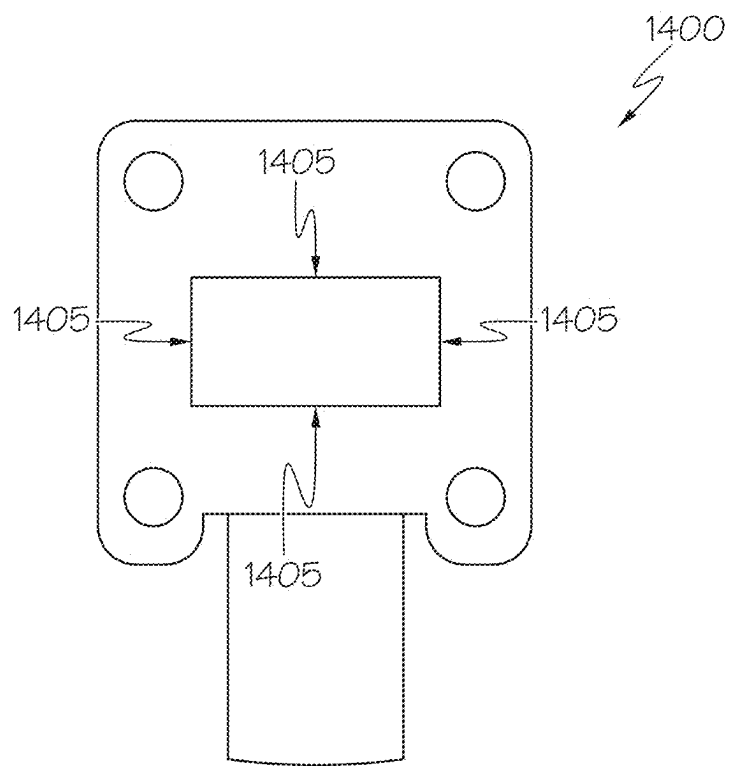


FIG. 14

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PHOTOPLETHYSMOGRAPHY SENSORS AND PROCESSES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/047,166, filed Oct. 17, 2022, entitled “PHOTOPLETHYSMOGRAPHY SENSORS AND PROCESSES,” which is a continuation of U.S. patent application Ser. No. 17/821,990, filed Aug. 24, 2022, entitled “PHOTOPLETHYSMOGRAPHY SENSORS AND PROCESSES,” which claims priority to U.S. Provisional Patent Application No. 63/236,511, filed Aug. 24, 2021, entitled “PHOTOPLETHYSMOGRAPHY SENSORS AND PROCESSES,” each of which are incorporated herein by reference in their entireties.

GOVERNMENT LICENSE RIGHTS

This invention was made with government support under NSF Award ID 2136470 awarded by the National Science Foundation. The government has certain rights in the invention.

BACKGROUND

Photoplethysmography (PPG) generally refers to light-based techniques for detecting volumetric changes in blood. Typically, PPG is performed by directing an input light source at a subject's skin, measuring an amount of light reflected and/or transmitted by the skin, and determining a biological signal (e.g., heart rate variance, dissolved oxygen, etc.) based on the amount of light measured. Previous approaches to performing PPG have relied on large sensing units that include multiple arrays of light-emitting elements (e.g., LEDs) and light-detecting elements (e.g., photodiodes). Such systems typically rely on large sensing footprints and redundant sensing elements to compensate for natural and motion-induced variations in skin surface especially for reflectance PPG. For example, a sleeping subject rotates from a first side to a second side, thereby shifting and warping skin surfaces and, thus, changing the light reflection and transmission behavior of the skin. In this example, past approaches may attempt to accommodate changes in skin surface by measuring signals over a large surface area via a multitude of PPG arrays.

Consequently, previous PPG sensing systems demonstrate large spatial profiles (e.g., due to sensing area requirements) and high complexity (e.g., due to multiple PPG arrays). Large spatial profiles and high complexity are undesirable for biological signal sensors and, especially so, for sensors intended to be actively worn by a subject. For example, highly complex PPG sensing systems may be cost prohibitive, difficult to calibrate, and unaccommodating of subject motion. Furthermore, PPG sensing systems with large spatial profiles may be unsuitable for everyday use outside of a clinical setting and/or for subjects with physical dimensions that fall outside of a typical or expected range.

Therefore, there exists a long felt but unmet need for motion-tolerant, low profile PPG sensing systems.

BRIEF DESCRIPTION OF THE DISCLOSURE

There are various factors that may influence quality and/or repeatability of PPG signals of a device worn on a patient's chest (and other locations). These factors include,

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but may not be limited to: 1) skin axial strain; 2) skin shear strain; 3) skin bending (or folding/buckling); 4) PPG sensor tilting; 5) skin contact (or potential loss of skin contact); 6) anatomy (e.g., bone, muscle, fat, arterial, venous, hair, body mass index (BMI), etc.); 7) blood pressure; 8) melanin pigmentation; and 9) local perfusion state. As will be discussed, various embodiments of the systems and processes discussed herein mitigate these potential factors by at least: a) strain isolation of the PPG sensor with a large (relative to the PPG sensor), rigid enclosure (helps mitigate skin strain and bending factors); b) conformality (elastic spacer-helps maintain constant skin contact); and c) load (helps maintain skin contact even with PPG sensor tilting and overcomes anatomical issues by applying pressure that matches or exceeds diastolic pressure to minimize the effect of venous or deoxygenated blood content).

According to a first aspect, a patch including: a silicon-based adhesive layer defining a silicon-based adhesive layer opening and including a first side and a second side; an acrylic-based adhesive layer connected to the first side of the silicon-based adhesive layer, wherein the acrylic-based adhesive layer is exposed on the second side of the silicon-based adhesive layer through the opening; a rigid support structure including a bottom surface area defining a rigid support structure opening for a photoplethysmography (PPG) system; and the PPG system extending through the rigid support structure opening and including: a first side and a second side; and a spacer operatively connected to the first side of the PPG sensor, wherein when the patch is attached to a patient's skin: the silicon-based adhesive layer attaches to a first portion of the patient's skin; the acrylic-based adhesive layer attaches to a second portion of the patient's skin, the second portion of the patient's skin corresponding to at least a portion of the bottom surface area of the rigid support structure extending through the silicon-based adhesive layer opening; and the acrylic-based adhesive layer attaches to a third portion of the patient's skin, the third portion of the patient's skin corresponding to an area of the acrylic-based adhesive layer in contact with the PPG sensor.

According to a second aspect, the patch of the first aspect or any other aspect, wherein the patch includes a vertical distance between the second side of the PPG sensor and the bottom surface area of the rigid support structure.

According to a third aspect, the patch of the second aspect or any other aspect, wherein, when attached to the patient's skin, the second side of the PPG sensor is pressed into the patient's skin.

According to a fourth aspect, the patch of the third aspect or any other aspect, wherein an amount of pressure exerted by the patch on the patient when the PPG sensor is pressed into the patient's skin is at least partially based on the vertical distance.

According to a fifth aspect, the patch of the fourth aspect or any other aspect, wherein the vertical distance is 10 mm or less.

According to a sixth aspect, the patch of the first aspect or any other aspect, wherein the spacer is elastic.

According to a seventh aspect, the patch of the sixth aspect or any other aspect, wherein the spacer is silicone.

According to an eighth aspect, the patch of the seventh aspect or any other aspect, wherein the spacer is porous.

According to a ninth aspect, the patch of the first aspect or any other aspect, further including a flexible printed circuit board (PCB) connected to first side of PPG sensor.

According to a tenth aspect, the patch of the ninth aspect or any other aspect, wherein the PCB collects signal data from the PPG sensor.

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According to an eleventh aspect, the patch of the tenth aspect or any other aspect, wherein the patch further transmits data from the PPG sensor to a computing system.

According to a twelfth aspect, a patch including: a first adhesive area; a second adhesive area adjacent to the first adhesive area; a rigid support structure including a bottom surface; a strain isolation area including at least a portion of the second adhesive area supported by the rigid support structure; and a photoplethysmography (PPG) sensor operatively connected to the rigid support structure including: a first side and a second side; and an elastic spacer operatively connected to the first side of the PPG sensor; a vertical distance between the bottom surface of the rigid support structure and the second side of the PPG sensor; a PPG area including the second side of the PPG sensor and a portion of the second adhesive area, wherein: the PPG sensor tilts with respect to the rigid support structure via the elastic spacer; and the strain isolation area surrounds the PPG area.

According to a thirteenth aspect, the patch of the twelfth aspect or any other aspect, wherein the PPG sensor is pressed into a patient's skin based at least in part on the vertical distance between the bottom surface of the rigid support structure and the second side of the PPG sensor.

According to a fourteenth aspect, the patch of the thirteenth aspect or any other aspect, wherein the amount of pressure exerted by the patch on the patient when the PPG sensor is pressed into the patient's skin is at least partially based on the vertical distance.

According to a fifteenth aspect, the patch of the fourteenth aspect or any other aspect, wherein the vertical distance is 10 mm or less.

According to a sixteenth aspect, the patch of the fifteenth aspect or any other aspect, wherein the PPG sensor tilts with respect to the rigid support structure via the elastic spacer, thereby creating conformal contact between the second side of the PPG sensor and the patient's skin when the PPG sensor is pressed into the patient's skin.

According to a seventeenth aspect, the patch of the sixteenth aspect or any other aspect, wherein: the patch includes a first adhesive layer; and the first adhesive area includes at least a portion of an adhesive side of the first adhesive layer.

According to an eighteenth aspect, the patch of the seventeenth aspect or any other aspect, wherein: the patch includes a second adhesive layer; and the second adhesive area includes at least a portion of an adhesive side of the second adhesive layer.

According to a nineteenth aspect, the patch of the eighteenth aspect or any other aspect, wherein: the second adhesive layer includes a non-adhesive side; and the second side of the PPG sensor contacts the non-adhesive side of the second adhesive layer.

According to a twentieth aspect, the patch of the nineteenth aspect or any other aspect, wherein: the PPG sensor extends through an opening formed by the bottom surface of the rigid support structure; and the elastic spacer is operatively connected to the rigid support structure.

According to a twenty-first aspect, patch for obtaining PPG signals from low perfusion anatomical sites including: a first adhesive layer defining an opening; a photoplethysmography (PPG) system including: a PPG sensor including a first side and a second side; a spacer operatively connected to a first side of the PPG sensor; a second adhesive layer in contact with the second side of the PPG sensor, the second adhesive layer and the second side of the PPG sensor extending through the opening; and a distance between an

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edge of the opening and the PPG sensor extending there-through, wherein the PPG sensor is strain isolated based at least in part on distance.

According to a twenty-second aspect, the patch of the twenty-first aspect or any other aspect, wherein the patch further includes a vertical distance between a surface of the first adhesive layer and the second side of the PPG sensor.

According to a twenty-third aspect, the patch of the twenty-second aspect or any other aspect, wherein: the patch is for attaching to a patient's skin via the first adhesive layer and the second adhesive layer; and the patch presses the second side of the PPG sensor into the patient's skin based at least in part on the vertical distance.

According to a twenty-fourth aspect, the patch of the twenty-third aspect or any other aspect, wherein the spacer is elastic.

According to a twenty-fifth aspect, the patch of the twenty-third aspect or any other aspect, wherein: the patch includes a support structure operatively connected to the spacer; and the spacer enables the PPG sensor to tilt with respect to the support structure, creating conformal contact with the patient's skin.

According to a twenty-sixth aspect, the patch of the twenty-fifth aspect or any other aspect, wherein the spacer is porous.

According to a twenty-seventh aspect, the patch of the twenty-sixth aspect or any other aspect, wherein the second adhesive layer is stronger than the first adhesive layer.

According to a twenty-eighth aspect, the patch of the twenty-seventh aspect or any other aspect, wherein the first adhesive layer is a silicon-based adhesive layer.

According to a twenty-ninth aspect, the patch of the twenty-eighth aspect or any other aspect, wherein the second adhesive layer is an acrylic-based adhesive layer.

According to a thirtieth aspect, the patch of the twenty-ninth aspect or any other aspect, or of any other claim, wherein the patch presses the second side of the PPG sensor into the patient's skin based at least in part on the vertical distance at a pressure of about 80-120 mm Hg, thereby minimizing effects of venous or deoxygenated blood content on PPG readings.

According to a thirty-first aspect, a patch including: a strain isolation zone including a first layer and a rigid support structure, the strain isolation zone surrounding a photoplethysmography (PPG) zone, wherein the first layer includes an adhesive side and a non-adhesive side; the PPG zone including the first layer and a PPG sensor, wherein the PPG sensor is in contact with the non-adhesive side of the first layer; and a contact zone surrounding the strain isolation zone and including a second layer, the second layer forming an opening thereby exposing the first layer and the PPG sensor therethrough.

According to a thirty-second aspect, the patch of the thirty-first aspect or any other aspect, wherein the second layer includes an adhesive side.

According to a thirty-third aspect, the patch of the thirty-second aspect or any other aspect, wherein the second layer adhesive side includes a weaker adhesive than the first layer adhesive side.

According to a thirty-fourth aspect, the patch of the thirty-third aspect or any other aspect, wherein the patch is for attaching to a patient's skin via the second layer adhesive side and the first layer adhesive side.

According to a thirty-fifth aspect, the patch of the thirty-fourth aspect or any other aspect, wherein the patch includes a vertical distance between the PPG sensor and the second layer adhesive side.

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According to a forty-first aspect, a patch including: a strain isolation zone including a strain isolation adhesive and a rigid support structure, the strain isolation zone surrounding a photoplethysmography (PPG) zone; the PPG zone including a PPG sensor and a PPG adhesive; and a contact zone surrounding the strain isolation zone and including a contact zone adhesive, the contact zone adhesive forming an opening thereby exposing the strain isolation zone and the PPG zone.

According to a forty-second aspect, the patch of the forty-first aspect or any other aspect, wherein the patch is configured for attaching to skin of a patient.

According to a forty-third aspect, the patch of the forty-second aspect or any other aspect, wherein the patch includes a spacer operatively connected to the rigid support structure and the PPG sensor.

According to a forty-fourth aspect, the patch of the forty-third aspect or any other aspect, wherein the spacer is elastic.

According to a forty-fifth aspect, the patch of the forty-fourth aspect or any other aspect, wherein the patch includes a vertical distance between a first end of the PPG sensor and the contact zone adhesive.

According to a forty-sixth aspect, the patch of the forty-fifth aspect or any other aspect, wherein the PPG zone exerts a force into the skin of the patient based at least in part on the vertical distance.

According to a forty-seventh aspect, the patch of the forty-sixth aspect or any other aspect, wherein the strain isolation zone pulls the skin of the patient towards the rigid support structure.

According to a forty-eighth aspect, the patch of the forty-seventh aspect or any other aspect, wherein the strain isolation adhesive is a stronger adhesive than the contact zone adhesive.

According to a forty-ninth aspect, the patch of the forty-eighth aspect or any other aspect, wherein the patch includes a first layer including the strain isolation adhesive and the PPG adhesive.

According to a fiftieth aspect, the patch of the forty-ninth aspect or any other aspect, wherein the patch includes a second layer including the contact layer adhesive.

According to a fifty-first aspect, the patch of the fiftieth aspect or any other aspect, wherein: the first layer is connected to the second layer; and the first layer and the first end of the PPG sensor extend through the opening.

According to a sixty-first aspect, a process for obtaining PPG data including: creating a strain isolation zone on a patient's skin by attaching a strain isolation adhesive backed by a rigid support structure to the patient's skin, the strain isolation zone surrounding a photoplethysmography (PPG) zone; creating the PPG zone by attaching a PPG adhesive to the patient's skin, the PPG adhesive backed by a PPG sensor; and creating a contact zone surrounding the strain isolation zone by attaching a contact zone adhesive to the patient's skin, the contact zone adhesive surrounding the strain isolation zone and the PPG zone.

According to a sixty-second aspect, the process of the sixty-first aspect or any other aspect, wherein the strain isolation adhesive is a stronger adhesive than the contact zone adhesive.

According to a sixty-third aspect, the process of the sixty-second aspect or any other aspect, wherein the patch includes a first layer including the strain isolation adhesive and the PPG adhesive.

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According to a sixty-fourth aspect, the process of the sixty-third aspect or any other aspect, wherein the patch includes a second layer including the contact layer adhesive.

According to a sixty-fifth aspect, the process of the sixty-fourth aspect or any other aspect, wherein the first layer is connected to the second layer.

According to a sixty-sixth aspect, the process of the sixty-fifth aspect or any other aspect, wherein the first layer and at least a portion of the PPG sensor extend through an opening formed by the second layer.

According to a sixty-seventh aspect, the process of the sixty-fifth aspect or any other aspect, wherein the patch includes a spacer operatively connected to the PPG sensor.

According to a sixty-eighth aspect, the process of the sixty-seventh aspect or any other aspect, wherein the spacer is elastic.

According to a sixty-ninth aspect, the process of the sixty-first aspect or any other aspect, wherein the patch includes a vertical distance between a first end of the PPG sensor and the contact zone adhesive.

According to a seventieth aspect, the process of the sixty-ninth aspect or any other aspect, wherein the PPG zone exerts a force into the skin of the patient based at least in part on the vertical distance.

According to a seventy-first aspect, the process of the seventieth aspect or any other aspect, wherein the strain isolation zone pulls the skin of the patient towards the rigid support structure. According to a seventy-second aspect, a patch or process of any other aspect, wherein: the PPG sensor is a first PPG sensor; the rigid support structure opening is a first rigid support structure opening; the rigid support structure includes one or more second support structure openings; the patch includes: one or more second PPG sensors; and a force sensor associated with the first PPG sensor and/or the one or more second PPG sensors; the one or more second PPG sensors extend through the one or more second support structure openings; each of the first PPG sensor and the one or more second PPG sensors exert a different force on a patient's skin.

Third-party systems fail to provide the ability to observe and monitor a variety of subject data points. Additionally, they fail to provide a single device attached at one location on a subject to observe each of these data points. The present disclosure provides for a single patch for monitoring and retrieving a plurality of data points from a subject. Additionally, the present disclosure provides a method and system for calculating and evaluating those various biometric data points while also detecting disorders of a subject associated with such data.

Using the processes and systems as disclosed, a provider may be able to easily identify issues associated with a subject and provide a diagnosis. Sleep apnea, and other breathing disorders, are common among subjects, and through the use of the processes and systems of the present disclosure, these disorders can be more easily and with more accuracy be diagnosed and treated.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates a first perspective view of an example patch device according to various embodiments of the present disclosure;

FIG. 1B illustrates a second perspective view of the example patch device according to various embodiments of the present disclosure;

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FIG. 1C illustrates a third perspective view of the example patch device according to various embodiments of the present disclosure;

FIG. 2 illustrates a first perspective view of a PPG sensor cradle according to various embodiments of the present disclosure;

FIG. 3 illustrates pressure effects of an exemplary PPG sensor/device on a patient's skin according to various embodiments of the present disclosure;

FIG. 4A illustrates a depth of a PPG sensor on a patient's skin according to various embodiments of the present disclosure;

FIG. 4B illustrates a pressure exerted by a PPG sensor on a patient's skin according to various embodiments of the present disclosure;

FIG. 4C illustrates a pressure exerted by a PPG sensor on a patient's skin according to various embodiments of the present disclosure;

FIG. 5 illustrates various strain zones according to various embodiments of the present disclosure;

FIG. 6A illustrates a first set of data points according to various embodiments of the present disclosure;

FIG. 6B illustrates a second set of data points according to various embodiments of the present disclosure;

FIG. 7A illustrates a third set of data points according to various embodiments of the present disclosure;

FIG. 7B illustrates a fourth set of data points according to various embodiments of the present disclosure;

FIG. 8A illustrates a fifth set of data points according to various embodiments of the present disclosure;

FIG. 8B illustrates a sixth set of data points according to various embodiments of the present disclosure;

FIG. 9 illustrates a first perspective view of a PPG sensor assembly according to various embodiments of the present disclosure;

FIG. 10 illustrates a cross-sectional view of the PPG sensor according to various embodiments of the present disclosure;

FIG. 11 illustrates a perspective view of a portion of the PPG sensor assembly according to various embodiments of the present disclosure;

FIG. 12 illustrates a perspective view of the PPG sensor assembly according to various embodiments of the present disclosure;

FIG. 13 illustrates a perspective view of the PPG sensor assembly according to various embodiments of the present disclosure; and

FIG. 14 illustrates a perspective view of a portion of the PPG sensor assembly according to various embodiments of the present disclosure.

While the disclosure is susceptible to various modifications and alternative forms, a specific embodiment thereof is shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but to the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

DETAILED DESCRIPTION

Aspects, features, and benefits of the systems, methods, processes, formulations, apparatuses, and products discussed herein will become apparent from the information disclosed herein and in the other application incorporated by

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reference. Variations and modifications to the disclosed systems and methods may be effected without departing from the spirit and scope of the novel concepts of the disclosure.

It will, nevertheless, be understood that no limitation of the scope of the disclosure is intended by the information disclosed herein or in the application incorporated by reference; any alterations and further modifications of the described or illustrated embodiments, and any further applications of the principles of the disclosure as illustrated therein are contemplated as would normally occur to one skilled in the art to which the disclosure relates.

According to various embodiments, aspects of the present disclosure generally relate to systems and apparatuses for deriving improved PPG signals. In one or more embodiments, the described PPG devices are configured to emit light against a subject's skin, measure a magnitude of light reflected from and/or transmitted through the subject's skin, and, based on the magnitude of light measured, determine one or more biological signals of the subject.

In one example, an embodiment of the present PPG device is affixed to a subject's skin and records reflected light intensities throughout the subject's sleep cycle. In the same example, based on the recorded intensities, the PPG device (or connected systems) determine, derive or estimate a variety of parameters and factors of the subject, including, but not limited to, the subject's dissolved oxygen (SpO₂) level, heart rate variability (HRV), pulse rate, and blood pressure, which may be used to monitor the subject's health and/or support evaluation of the subject for one or more conditions (e.g., sleep apnea and other sleep-related disorders). In one or more embodiments, the present PPG devices provide advantages including, but not limited to, low spatial profile, high tolerance to subject motion, greater signal amplitude on low perfusion anatomical sites, and compatibility with any skin tone or pigment.

For example, in contrast to previous approaches that require numerous and outsized light sensing arrays, the PPG solution presented herein may operate with a single PPG array that demonstrates a minimal spatial footprint. As another example, past approaches typically require redundant PPG sensing units to compensate for losses in signal quality at one or more PPG sensing elements during skin strain and deformation (e.g., as caused by subject motion, clothing, or external phenomena) or on low perfusion anatomical sites. In contrast, the present PPG solution may operate with a single PPG sensing unit that tolerates subject motion, clothing, and other external phenomena without experiencing significant losses in accuracy and precision. The present PPG solution may also function on low perfusion anatomical sites.

FIG. 1A illustrates a first perspective view of an example patch device according to various embodiments of the present disclosure.

According to various embodiments, the patch device includes a patch **100** and a pod **150**. In one embodiment, the patch **100** includes a cradle **105**. According to some embodiments, the pod **150** attaches to the patch **100** via a latch on the cradle **105**. As will be understood from discussions herein, the patch **100** and pod **150** may be one integrally formed device or may be split into more components than shown in FIG. 1A. As will also be understood, patch **100** and pod **150** may connect via any suitable mechanism not shown (e.g., a suitable adhesive or the like).

The cradle **105** may include an opening **110**. In various embodiments, the opening **110** is generally a rectangular shape with rounded corners. In at least one embodiment, the

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opening **110** is any suitable shape, such as, but not limited to, circular, ovoid, square, etc. According to one embodiment, the opening **110** receives a PPG sensor and may be shaped to receive the PPG sensor. In at least one embodiment, the opening is generally a square shape (with rounded corners) and each side of the square is about 16-20 mm.

The cradle **105** may be constructed of any suitable material. In at least one embodiment, the cradle **105** is constructed of a stiff material (or materials), including any suitable plastic material.

As shown in FIG. 1A, the patch **100** includes a second adhesive **130** (FIG. 1B), which spans the opening **110**. In these embodiments (and others), the patch **100** includes a top side **115** of a second adhesive layer **130** seen through opening **110** (as further described with reference to FIG. 1B). As will be understood, the second adhesive layer **130** (as seen through opening **110**) may be part of a larger adhesive layer that spans a portion of the patch **100** or may be an adhesive layer for covering opening **110** (and/or an additional small portion of the patch **100**). As will also be understood, the adhesive layer **130** may include a non-adhesive side shown in FIG. 1A and an adhesive side shown in FIG. 1B.

FIG. 1B illustrates a second perspective view of the example patch device according to various embodiments of the present disclosure. As shown in FIG. 1B, the patch **100** includes a first adhesive layer **125**, a second adhesive layer **130**, and two electrocardiogram (ECG) electrodes **120** on each side of the patch **100**.

In the embodiment shown, the first adhesive layer **125** spans substantially the entire bottom area of the patch **110**, except for areas that are cut out (or where the first adhesive layer **125** defines openings) for the ECG electrodes **120** and second adhesive layer **130**. As will be understood from discussions herein, the patch **100** may include any suitable number of adhesive layers or suitable openings/cut outs for different components, sensors, or features.

The second adhesive layer **130** may be any suitable size. As discussed above in relation to FIG. 1A, the second adhesive layer **130** may be sized to cover the opening **110**. In the embodiment shown in FIG. 1B, the second adhesive layer **130** may span a cut out/opening in the first adhesive layer, which may generally be the size of the cradle **105** (or be any suitable size).

In one embodiment, the second adhesive layer **130** includes a third adhesive area **520** of FIG. 5 (as discussed herein).

The first adhesive layer **125** and the second adhesive layer **130** may include adhesives of any strength. According to an embodiment, the first adhesive layer **125** is first strength. In at least one embodiment, the first adhesive layer **125** is an adhesive made of silicon. In some embodiments, the second adhesive layer **130** is a second strength. According to an embodiment, the second adhesive layer is made of acrylic. In some embodiments, the first adhesive layer **125** is made of a weaker adhesive than the second adhesive layer **135**. The adhesive layers discussed herein may be of any strengths. In one such embodiment, the first adhesive layer **125** and the second adhesive layer **130** are of the same or similar strengths. In further embodiments, the patch **100** includes fewer or more adhesive layers than discussed herein.

As shown in FIG. 1B, the pod **150** may include an elastic spacer **155** and a PPG sensor **160** (each of which are discussed further herein) extending from the pod **150**. In some embodiments, the elastic spacer **155** and the PPG sensor **160** are integrally formed with the pod **150**. In at least

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one embodiment, the elastic spacer **155** and the PPG sensor **160** are separate from the pod **150**, but are operatively connected to the same during assembly of the pod **150**. In one embodiment, the elastic spacer **155** and the PPG sensor **160** are connected to or attached to the cradle **105** or patch **100** (opposed to the pod **150**).

The pod **150** may include additional computing components not shown. In at least one embodiment, the pod **150** connects to the PPG sensor via a PPG flexible printed circuit board (as further discussed herein). In some embodiments, the pod **150** includes additional computing components to receive, transmit, interpret, filter, and/or analyze information received from the PPG sensor **160** (and/or ECG electrodes **120**). In one embodiment, the pod **150** includes a cellular, Bluetooth, or BLE antenna (or combination of antennas) for transmitting data received from the PPG sensor **160** to a computing system for analysis and/or diagnosis.

According to particular embodiments, the elastic spacer **155** and PPG sensor **160** extend a particular distance from the bottom of the pod **150**. In at least one embodiment, the elastic spacer and the PPG sensor **160** extend approximately 0.1-10.0 mm from the bottom of the cradle **105**. In some embodiments the elastic spacer **155** and the PPG sensor **160** extend less than 6.1 mm from the bottom of the pod **150**.

FIG. 1C illustrates a third perspective view of the example patch device according to various embodiments of the present disclosure. In this embodiment, the pod **150** is attached to the patch **100** via the cradle **105**. When the pod **150** is attached to the patch **100**, the PPG sensor **160** may push on the second adhesive layer **130**, creating a protrusion **135** (e.g., the elastic spacer **155** and the PPG sensor **160** extend through the opening **115** in the cradle **105**, press on a non-adhesive side of the second adhesive layer **130**, and deform the second adhesive layer **130**). As will be further discussed in relation to FIG. 5, the various bottom features of the assembled pod **150** and patch **100** form various areas or zones. In a particular embodiment, the area of the protrusion **135** may have certain effects on the PPG signal quality. In these embodiments (and others), the second adhesive **130** that is not part of the protrusion **135**, but is backed by the stiff cradle **105** and/or pod **150** forms a second area/zone (of strain isolation). In further embodiments, the area of the first adhesive **125** may form an additional/third area/zone.

FIG. 2 illustrates a first perspective view of a PPG sensor cradle according to various embodiments of the present disclosure. As will be understood from discussions herein, the PPG sensor cradle **205** illustrates various components shown in a particular form to illustrate various effects of this arrangement, but the components shown may appear differently in other embodiments (e.g., as shown in FIGS. 1A-1C).

According to various embodiments and as illustrated in FIG. 2, the PPG sensor cradle **205** sits on top of a subject's skin **210** and is attached to the same via an adhesive layer **245** (which may correspond to second adhesive layer **130** shown in FIGS. 1A-1C).

The PPG sensor cradle **205** includes an enclosure **215**. In one embodiment, the enclosure **215** houses various PPG sensor components. According to various embodiments, the enclosure **215** is rigid and may be made of various materials, including any suitable plastic or like rigid and lightweight material. The rigid enclosure may provide for additional pressure to be applied to a subject, as opposed to a flexible enclosure.

Additionally, the enclosure **215** may be any shape, including a trapezoidal shape as shown in FIG. 1.

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According to one embodiment, the enclosure **215** includes a PPG sensor **235**, a spacer **240**, and an elastic spacer **225**. As shown in FIG. 2, the PPG sensor **235** extends beyond the adhesive layer **245** and presses into the subject's skin **210** via the spacer **240** and elastic spacer **220**. In the embodiment shown in FIGS. 1A-1C, the second adhesive layer **130** (e.g., adhesive layer **245** in FIG. 2), extends over a protruding PPG sensor **160**. It will be understood that in various embodiments, the adhesive **245** may extend over the PPG sensor **235** as shown in FIG. 2 and include the same general effects and features as discussed herein.

In one embodiment, the elastic spacer **220** sits within the enclosure **215**. As shown in the embodiment in FIGS. 1A-1C, the elastic spacer **220** may extend from a bottom of an enclosure **215** (e.g., pod **150** and/or cradle **105**).

In one or more embodiments, when the enclosure **215** is attached to the subject's skin **210** (via the adhesive **245**), the adhesive and the elastic spacer **220** (and the spacer **240**) apply a load to the PPG sensor **235** in the direction of the subject's skin **210**. The elastic spacer **220** may be made of various elastic materials, including, for example, 2A silicone, rubber, or other flexible materials. According to various embodiments, the elastic spacer **220** includes a porous architecture. For example, the spacer **220** may include 2A silicone arranged into a lattice structure including a plurality of voids.

As will be understood from discussions herein, the elastic spacer **220** may enable the PPG sensor **235** to be in conformal contact with the subject's skin **210** (e.g., the PPG sensor **235** can tilt/move with the subject's skin **210** via deformation/compression of the elastic spacer **220**). In some embodiment, this allows a PPG sensor **235** (discussed further below) to continue reading data from a subject, even when the subject rolls or breathes.

According to one embodiment, the PPG sensor cradle includes a gap **230** measuring from the first adhesive layer **245** and the bottom of the PPG sensor **235**. According to one embodiment, this gap **230** may provide a backing force applied on the PPG sensor **235**. This backing force may enhance signal quality of the readings from the PPG sensor **235**. In various embodiments, the applied load biases the PPG sensor **235** toward the skin and ensures the PPG sensor **235** remains in conformity with the skin regardless of subject motion or other dislodging phenomena. As will be understood from discussions herein, the gap **230** may be created by any arrangement and number of components (e.g., spacers, portions of a cradle, portions of a patch, etc.). In at least one embodiment, the gap **230** may be about 0.1 mm, 0.2 mm, 0.1-2.0 mm, 1.1 mm or less, 0.1-10 mm, 6.1 mm or less, or any suitable distance.

According to one embodiment, a combination of features of the PPG cradle **205** may isolate a region from strain (e.g., create a strain isolation area). In at least one embodiment, isolation from strain causes the region of skin to remain taut (e.g., unfolded, unwrinkled, etc.) regardless of the subject's motion and despite any warps or other changes to skin surrounding the region of strain isolation. As shown in FIG. 2, the gap **230** and force applied by the various components shown therein at least partially causes skin **210** directly under a portion of the PPG cradle **205** surround the elastic spacer **220** to pull the subject's skin **210** upwards toward the PPG cradle **205**, creating an area of strain isolation.

In one or more embodiments, the combination of the strain isolation and continuous conformity between the PPG sensor **235** and the skin provides for more accurate and precise PPG sensor readings.

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FIG. 3 shows pressure effects on a patient's skin from an exemplary PPG sensor/device, according to various embodiments of the present disclosure.

According to at least one embodiment, the PPG sensor cradle **205** may be in contact with a subject's skin **210**. In various embodiments, an amount of force applied relates to the gap **230** as discussed with reference to FIG. 2. The force may be created due to the gap **230** and the adhesives applied to a subject's skin. Additionally, in one embodiment, the subject's body mass index (BMI) impacts the amount of force applied.

For example, and as illustrated in FIG. 3, one (1) Newton (N) of force is applied to the PPG sensor cradle **205**. This 1N of force may be created from the gap between the first adhesive layer **245** and the bottom of the PPG sensor **235**, as discussed with reference to FIG. 2. As illustrated, the force may differ depending on the size of the gap **230** and the subject's BMI. As shown, the patient's skin and underlying tissue experiences varying levels of pressure (expressed in mmHg). As shown in FIGS. 8A and 8B, increasing force/load may increase PPG signal amplitude.

As will be understood, maintaining force and skin conformality with the PPG sensor and patient's skin may lead to higher levels of repeatability. Further, in one or more embodiments, the gap **230** and force applied by the various components shown herein applies a pressure to the tissue under skin **210** directly under a portion of the PPG cradle **205** as shown in FIG. 3 that matches or exceeds diastolic pressure (e.g., about 80 to 120 mmHg) to minimize the effect of venous or deoxygenated blood content that provides more accurate and precise PPG sensor readings.

FIG. 4A illustrates a view of the PPG sensor cradle according to various embodiments of the present disclosure. FIG. 4A demonstrates maintaining a constant depth (shown in mm) that can be achieved using the systems and methods here (e.g., spacers, gaps, rigid enclosures). FIG. 4B illustrates a view of the PPG sensor cradle according to various embodiments of the present disclosure. FIG. 4B shows that a relatively consistent pressure (in kPa) is achieved through the techniques and devices discussed herein even when a patient's skin tilts or rocks (e.g., when a patient moves). FIG. 4C illustrates a view of the PPG sensor cradle according to various embodiments of the present disclosure. FIG. 4C shows that consistent pressure (in kPa) is maintained by the techniques and devices herein when a patient rolls (e.g., during a sleep test).

According to one embodiment, the PPG sensor cradle **205** may sit at different angles. As illustrated by a first view **405** of the PPG sensor cradle **205**, the PPG sensor cradle **205** may sit flat. As shown in FIGS. 4B and 4C, even as a patient rotates, in various embodiments, the system/device shown herein maintains a relatively consistent pressure on the patient's skin, thus resulting in relatively consistent PPG readings/signals.

As illustrated by a second view **410** and third view **415**, the PPG sensor cradle **205** may sit at an angle. For example, if the PPG sensor cradle **205** is attached to a subject, the subject may breathe or roll and adjust the angle of chest while keeping the PPG sensor cradle attached and capable of adjusting in accordance with such breathing.

Moreover, depending on the amount of force applied to the PPG sensor cradle **205**, the PPG sensor cradle **205** may be prevented from rocking. As discussed with reference to FIG. 3, when 1N of force is applied, the PPG sensor cradle **205** is pressed onto a subject's skin. Thus, the increased pressure leads to decreased rocking (if any) of the PPG sensor cradle **205**. The increased pressure of the underlying

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tissue can also match or exceed diastolic pressure to remove venous or deoxygenated blood content from data derived from the PPG sensor to provide more accurate and precise PPG sensor readings.

FIG. 5 illustrates various strain zones according to particular embodiments of the present disclosure. FIG. 5 shows zones on a patient's skin and the effects of different corresponding areas of various embodiments of devices/systems discussed here.

In one embodiment, FIG. 5 includes an affected area 505, which may correspond to the example patch device 105 in accordance with FIG. 1. According to various embodiments, the affected area 505 may include a first adhesive area 510, a second adhesive area 515, and a third adhesive area 520. In one embodiment, the first adhesive area 510 may correspond to the first adhesive layer 125 of FIGS. 1A-1C (FIG. 2 does not include an area corresponding to first adhesive layer 125 of FIGS. 1A-1C). According to one embodiment, the second adhesive area 515 may correspond to the second adhesive layer 130 of FIG. 1 and adhesive layer 245 of FIG. 2. The second adhesive area 515 may be a stronger adhesive than the first adhesive area 510. In one embodiment, the third adhesive area 520 may be made of the same adhesive as the second adhesive area 515. According to one embodiment, the third adhesive area 520 may correspond to the second adhesive layer 245 plus the PPG sensor 235 of FIG. 2 or the second adhesive layer 130 and the protrusion 135 of FIGS. 1A-1C.

In one embodiment, the weaker first adhesive area 510 prevents a subject from irritation or bruising where the adhesive attaches to the subject's skin. However, because the first adhesive area 510 is weaker, the subject's skin may move.

According to various embodiments, the stronger second adhesive area 515 is backed by the enclosure 215 of FIG. 2 or the cradle 105 (and/or pod 150) of FIG. 1. In at least one embodiment, the rigid backing leads to increased strength of attachment of the adhesive and the subject's skin and helps prevent irritation or bruising, despite a stronger adhesive. According to particular embodiments, the second adhesive area 515 is an area of strain isolation as discussed herein.

In various embodiments, the stronger third adhesive area 520 sits below the PPG sensor 235 of FIG. 2. However, the PPG sensor 235 is not directly attached to the third adhesive area 520. Rather, the PPG sensor 235 sits behind (from the perspective of a patient's skin) the third adhesive area 520; this allows for the PPG sensor 235 to move freely behind the third adhesive area 520, leading to conformality of the sensor and increased accuracy of data collected from the subject. In one embodiment, force is applied to the third adhesive area 520, as discussed above with reference to FIG. 3.

FIG. 6A illustrates a first set of data points according to various embodiments of the present disclosure. FIG. 6B illustrates a second set of data points according to various embodiments of the present disclosure. According to various embodiments, graphs 600 and 650 illustrate the correlation between the PPG attachment mechanism and the quality of various PPG data points collected from a subject. FIGS. 6A and 6B show a quality of signal for PPG red (PPGr) and infrared (PPGi) signals based on three different attachment mechanisms: 1) overtape (e.g., PPG sensor is taped to patient's chest); 2) rigid placement method (e.g., PPG sensor is attached to patient's chest via a rigid structure (e.g., a PPG cradle or like structure discussed herein); and 3) 3D printed elastic (e.g., the combination of a rigid structure and an elastic spacer as discussed herein).

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Graph 600 illustrates the impact of placement method on the quality of PPGr. At block 605, overtape is used for placement of the PPG sensor. The use of overtape has a lower quality of PPGr than the other two placement mechanisms. As shown in FIG. 6A, PPGr signal quality using overtape is about 0.025 to 0.125.

At block 610, a rigid placement of the PPG sensor is used. The rigid placement leads to increased quality of the PPGr relative to the use of overtape. As shown in FIG. 6A, PPGr signal quality using a rigid placement method is about 0.075 to 0.200.

At block 615, 3D printed elastic is used (e.g., in combination with a rigid structure). In the embodiment shown, the use of an elastic spacer in combination with a rigid structure greatly increases the quality of PPGr relative to both the overtape and rigid placement because the PPG sensor is conformally attached to the patient (e.g., as discussed herein, the PPG sensor can tilt/rock/move via compression of the elastic spacer to maintain strong contact with a patient's skin). As shown in FIG. 6A, use of the 3D printed elastic mechanism results in a PPGr quality of about 0.300 to 0.500.

Graph 650 illustrates the impact of placement method on the quality of PPGi. At block 655, overtape is used for placement of the PPG sensor. The use of overtape has a lower quality of PPGi than the other two placement mechanisms. As shown in FIG. 6B, PPGi quality when using an overtape mechanism is about 0.025 to 0.160.

At block 660, a rigid placement of the PPG sensor is used. The rigid placement leads to increased quality of the PPGi relative to the use of overtape. As shown in FIG. 6B, PPGi quality when using a rigid placement mechanism is about 0.085 to 0.250.

At block 665, 3D printed elastic is used in the placement of the PPG sensor (in combination with a rigid structure). The use of an elastic spacer greatly increases the quality of PPGi relative to both the overtape and rigid placement. As shown in FIG. 6B, PPGi quality when using 3D printed elastic is about 0.400 to 0.610.

FIG. 7A illustrates a third set of data points according to various embodiments of the present disclosure. FIG. 7B illustrates a fourth set of data points according to various embodiments of the present disclosure.

According to various embodiments, graphs 700 and 750 illustrate a correlation between the enclosure method of a PPG sensor and the quality of various PPG data points collected from a subject. As shown in graphs 700 and 750, a number of enclosure environments were tested: 1) no enclosure (e.g., PPG sensor just taped or otherwise attached to a patient); 2) a rigid enclosure (e.g., a relatively small footprint "enclosure" or rigid backing); and 3) a strain isolating rigid enclosure (e.g., an enclosure with a larger footprint that surrounds the PPG sensor and any spacers, effectively strain isolating the PPG sensor and creating the various zones shown in FIG. 5).

Graph 700 illustrates the impact of the enclosure method on the quality of PPGr. At block 705, no enclosure is used. The use of no enclosure leads to a lower quality of PPGr than other enclosure mechanisms. As shown in FIG. 7A, no enclosure results in a PPGr of about 0.025 to 0.125.

At block 710, an enclosure is used. The use of the enclosure leads to increased quality of the PPGr relative to the use of no enclosure. As shown in FIG. 7B, a rigid (but not a strain isolating enclosure) results in a PPGr of about 0.085 to 0.425.

At block 715, strain isolation is used. The use of strain isolation greatly increases the quality of PPGr relative to both the use of no enclosure or the use of an enclosure. As

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shown in FIG. 7A, use of a strain isolating enclosure (such as those discussed herein, including as shown in FIGS. 1 and 2) results in a PPGi of about 0.185 to 0.525).

Graph 750 illustrates the impact of the enclosure method on PPGi. At block 755, no enclosure is used. The use of no enclosure leads to a lower quality of PPGi compared to other enclosure mechanisms. As shown in FIG. 7B, no enclosure results in PPGi of about 0.025 to 0.160.

At block 760, an enclosure is used (e.g., but not a strain isolating enclosure). The use of the enclosure leads to increased quality of the PPGi relative to the use of no enclosure. As shown in FIG. 7B, a non-strain isolating enclosure results in a PPGi of about 0.115 to 0.560.

At block 765, strain isolation is used. The use of strain isolation greatly increases the quality of PPGi relative to both the use of no enclosure or the use of an enclosure. As shown in FIG. 7B, a strain isolating enclosure results in a PPGi of about 0.250 to 0.630.

FIG. 8A illustrates a fifth set of data points according to various embodiments of the present disclosure. FIG. 8B illustrates a sixth set of data points according to various embodiments of the present disclosure.

According to various embodiments, the graphs 800 and 850 illustrate the correlation between the force applied to the PPG sensor and amplitude of red and infrared (IR) signals collected from a subject. Each of lines 805, 855, 810, 860, 815, 865, 820, and 870, correspond to different subjects or different tests. According to one embodiment, the various points on the line correspond to 0.5N, 1.2N, 3N, and 3.75N applied to the subject's skin via the PPG sensor cradle.

In one embodiment, the force applied may correspond to FIG. 3. As shown, when more force is applied to the PPG sensor (e.g., against the subject's skin), the amplitude red and amplitude IR increases in strength. Thus, as illustrated, the force may increase the amplitude of signals received from a PPG sensor.

FIG. 9 illustrates a first perspective view of a PPG sensor assembly according to various embodiments of the present disclosure. The PPG sensor shown in FIGS. 9-14 may correspond to the PPG sensor shown in FIGS. 1A-1C (e.g., shown as part of the pod 150).

In one embodiment, the PPG sensor may include a first silicone layer 905, a PPG flexible printed circuit board (PCB) 910, and a second silicone layer 915. In one embodiment, the silicone layer 905 corresponds to the spacer 155 of FIGS. 1A-1C. In another embodiment, 905 and 915 are a single silicone layer. FIG. 9 includes a cross-sectional line 10-10 indicating the cross-section shown in FIG. 10.

FIG. 10 illustrates a cross-sectional view of the PPG sensor according to various embodiments of the present disclosure. As shown, the PPG sensor may be viewed along the 10-10 cross section. This view of the PPG sensor shows exemplary internal components of the first silicone layer 905, a PPG printed circuit board 910, and a second silicone layer 915, as discussed with reference to FIG. 9. In the embodiment shown, the first silicone layer 905 is an elastic spacer with one or more pores or otherwise voided structure 1005. Further, in various embodiments, the PPG sensor includes PPG sensing components (e.g., red and IR components) shown at 1015, which may be encased in plastic or another protective barrier as shown (e.g., 1020). As also shown in the embodiment of FIG. 10, a portion of the PPG printed circuit board 1010 weaves or otherwise passes through the spacer/first silicone layer 905 to connect the PPG sensor to corresponding computing components included in a pod (e.g., pod 150).

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As will be understood from discussions herein, in various embodiments, the spacer/first silicone layer 905 may compress or otherwise deform such that the PPG sensor is in conformal contact with a patient's skin. In these embodiments (and others), the PPG printed circuit board 910 is flexible at least to withstand the compression of the spacer/first silicone layer 905. In one embodiment, particular portions of the PPG printed circuit board 910 are flexible (e.g., portion 1010). In further embodiments, the entire PPG printed circuit board 910 is flexible.

FIG. 11 illustrates a perspective view of a portion of the PPG sensor assembly according to various embodiments of the present disclosure. According to various embodiments, the PPG sensor may include a PPG printed circuit board 1105, which in one embodiment, may correspond to the PPG printed circuit board 910 of FIG. 9.

FIG. 12 illustrates a perspective view of the PPG sensor assembly according to various embodiments of the present disclosure. In the embodiment shown, the PPG sensor includes a support structure 1205 connected to the first silicone layer (e.g., elastic spacer) 905. In at least one embodiment, the support structure includes various holes/voids (or pores) 1210 that may result in weight reduction.

The support structure 1205 may be constructed of any suitable material. In at least one embodiment, the support structure is constructed of a hard plastic material and is used for operatively connecting the PPG sensor with a pod casing (or other suitable structure).

FIG. 13 illustrates a perspective view of the PPG sensor assembly according to various embodiments of the present disclosure.

FIG. 14 illustrates a perspective view of a portion of the PPG sensor assembly according to various embodiments of the present disclosure. According to one embodiment, this perspective view shows the top of the PPG sensor assembly and the top of the PPG printed circuit board 910 of FIG. 9. In various embodiments, the PPG printed circuit board 910 may include a gap-filling adhesive 1405.

The foregoing description of the exemplary embodiments has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the inventions to the precise forms disclosed. Many modifications and variations are possible in light of the above teaching.

One such variation includes a device/patch that includes a force sensor associated with a PPG sensor to measure the amount of force exerted on a patient via the patch. In these embodiments (and others), the force may be tunable via an elastic spacer backing the PPG sensor and/or a vertical distance between the PPG sensor and a bottom of a rigid structure (e.g., cradle).

In further embodiments, a patch includes force sensors and multiple PPG sensors. In these embodiments (and others), a rigid support structure includes openings for, or is otherwise operatively connected to, multiple PPG sensors with elastic or other backings (such as those PPG sensors with elastic spacers discussed herein). Continuing with this embodiment, each PPG sensor of the multiple PPG sensors exerts a different force (resulting in a different pressure) on a patient's skin. As will be understood, in the embodiments above, a system may receive data from the force sensors and multiple PPG sensors and be able to calculate or otherwise determine a patient's blood pressure.

In still further embodiments, a patch includes one or more PPG sensors (which may or may not be associated with or connected to an elastic spacer) associated with one or more force sensors, where the force exerted on a patient's skin

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may be tunable via an actuator, motor, or other mechanical mechanism for adjusting a vertical distance between the one or more PPG sensors and a portion of a rigid support structure. In alternate embodiments, the patch includes a spacer connected to at least one of the one or more PPG sensors, where the spacer degrades, is a shape memory polymer, or otherwise changes over time to decrease (or increase) a vertical distance, compressibility of the spacer (so the spacer compresses more or less), or otherwise changes the force on a patient's skin over time.

The embodiments were chosen and described in order to explain the principles of the inventions and their practical application so as to enable others skilled in the art to utilize the inventions and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present inventions pertain without departing from their spirit and scope. Accordingly, the scope of the present inventions is defined by the appended claims rather than the foregoing description and the exemplary embodiments described therein.

From the foregoing, it will be understood that various aspects of the processes described herein are software processes that execute on computer systems that form parts of the system. Accordingly, it will be understood that various embodiments of the system described herein are generally implemented as specially-configured computers including various computer hardware components and, in many cases, significant additional features as compared to conventional or known computers, processes, or the like, as discussed in greater detail herein. Embodiments within the scope of the present disclosure also include computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable media can be any available media which can be accessed by a computer, or downloadable through communication networks. By way of example, and not limitation, such computer-readable media can comprise various forms of data storage devices or media such as RAM, ROM, flash memory, EEPROM, CD-ROM, DVD, or other optical disk storage, magnetic disk storage, solid state drives (SSDs) or other data storage devices, any type of removable non-volatile memories such as secure digital (SD), flash memory, memory stick, etc., or any other medium which can be used to carry or store computer program code in the form of computer-executable instructions or data structures and which can be accessed by a computer.

When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computer, the computer properly views the connection as a computer-readable medium. Thus, any such a connection is properly termed and considered a computer-readable medium. Combinations of the above should also be included within the scope of computer-readable media. Computer-executable instructions comprise, for example, instructions and data which cause a computer to perform one specific function or a group of functions.

Those skilled in the art will understand the features and aspects of a suitable computing environment in which aspects of the disclosure may be implemented. Although not required, some of the embodiments of the claimed inventions may be described in the context of computer-executable instructions, such as program modules or engines, as described earlier, being executed by computers in networked environments. Such program modules are often reflected and illustrated by flow charts, sequence diagrams, exem-

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plary screen displays, and other techniques used by those skilled in the art to communicate how to make and use such computer program modules. Generally, program modules include routines, programs, functions, objects, components, data structures, application programming interface (API) calls to other computers whether local or remote, etc. that perform particular tasks or implement particular defined data types, within the computer. Computer-executable instructions, associated data structures and/or schemas, and program modules represent examples of the program code for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps.

Those skilled in the art will also appreciate that the claimed and/or described systems and methods may be practiced in network computing environments with many types of computer system configurations, including personal computers, smartphones, tablets, hand-held devices, multiprocessor systems, microprocessor-based or programmable consumer electronics, networked PCs, minicomputers, mainframe computers, and the like. Embodiments of the claimed invention are practiced in distributed computing environments where tasks are performed by local and remote processing devices that are linked (either by hardwired links, wireless links, or by a combination of hardwired or wireless links) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices.

An exemplary system for implementing various aspects of the described operations, which is not illustrated, includes a computing device including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. The computer will typically include one or more data storage devices for reading data from and writing data to. The data storage devices provide nonvolatile storage of computer-executable instructions, data structures, program modules, and other data for the computer.

Computer program code that implements the functionality described herein typically comprises one or more program modules that may be stored on a data storage device. This program code, as is known to those skilled in the art, usually includes an operating system, one or more application programs, other program modules, and program data. A user may enter commands and information into the computer through keyboard, touch screen, pointing device, a script containing computer program code written in a scripting language or other input devices (not shown), such as a microphone, etc. These and other input devices are often connected to the processing unit through known electrical, optical, or wireless connections.

The computer that effects many aspects of the described processes will typically operate in a networked environment using logical connections to one or more remote computers or data sources, which are described further below. Remote computers may be another personal computer, a server, a router, a network PC, a peer device or other common network node, and typically include many or all of the elements described above relative to the main computer system in which the inventions are embodied. The logical connections between computers include a local area network (LAN), a wide area network (WAN), virtual networks (WAN or LAN), and wireless LANs (WLAN) that are presented here by way of example and not limitation. Such networking environments are commonplace in office-wide or enterprise-wide computer networks, intranets, and the Internet.

When used in a LAN or WLAN networking environment, a computer system implementing aspects of the invention is connected to the local network through a network interface or adapter. When used in a WAN or WLAN networking environment, the computer may include a modem, a wireless link, or other mechanisms for establishing communications over the wide area network, such as the Internet. In a networked environment, program modules depicted relative to the computer, or portions thereof, may be stored in a remote data storage device. It will be appreciated that the network connections described or shown are exemplary and other mechanisms of establishing communications over wide area networks or the Internet may be used.

While various aspects have been described in the context of a preferred embodiment, additional aspects, features, and methodologies of the claimed inventions will be readily discernible from the description herein, by those of ordinary skill in the art. Many embodiments and adaptations of the disclosure and claimed inventions other than those herein described, as well as many variations, modifications, and equivalent arrangements and methodologies, will be apparent from or reasonably suggested by the disclosure and the foregoing description thereof, without departing from the substance or scope of the claims. Furthermore, any sequence(s) and/or temporal order of steps of various processes described and claimed herein are those considered to be the best mode contemplated for carrying out the claimed inventions. It should also be understood that, although steps of various processes may be shown and described as being in a preferred sequence or temporal order, the steps of any such processes are not limited to being carried out in any particular sequence or order, absent a specific indication of such to achieve a particular intended result. In most cases, the steps of such processes may be carried out in a variety of different sequences and orders, while still falling within the scope of the claimed inventions. In addition, some steps may be carried out simultaneously, contemporaneously, or in synchronization with other steps.

The embodiments were chosen and described in order to explain the principles of the claimed inventions and their practical application so as to enable others skilled in the art to utilize the inventions and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the claimed inventions pertain without departing from their spirit and scope. Accordingly, the scope of the claimed inventions is defined by the appended claims rather than the foregoing description and the exemplary embodiments described therein.

Any logic or application described herein may be implemented and structured in a variety of ways. For example, one or more applications described may be implemented as modules or components of a single application. Further, one or more applications described herein may be executed in shared or separate computing systems or a combination thereof. For example, more than one application may execute in the same computing system, or in multiple computing systems in the same computing environment. Additionally, it is understood that terms such as "application," "service," "system," "engine," "module," and so on may be interchangeable and are not intended to be limiting.

Disjunctive language such as the phrase "at least one of X, Y, or Z," unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should

not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

It should be emphasized that the above-described embodiments of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

The invention claimed is:

1. A patch comprising:

a silicone-based adhesive layer;

an acrylic-based adhesive layer;

a rigid support structure comprising:

an enclosure for housing a compressible spacer; and

a bottom surface area defining a rigid support structure opening for a photoplethysmography (PPG) sensor; and

the PPG sensor comprising:

a first side and a second side, the first side opposite the second side; and

the compressible spacer located within the enclosure and comprising a top portion operatively connected to a top surface of the enclosure opposite the bottom surface area and a bottom portion operatively connected to the first side of the PPG sensor, wherein when the patch is attached to a patient's skin:

the second side of the PPG sensor is in contact with the patient's skin;

the compressible spacer is between the first side of the PPG sensor and the rigid support structure within the enclosure such that the top portion of the compressible spacer is operatively connected to the top surface of the enclosure and opposite the bottom portion of the compressible spacer connected to the first side of the PPG sensor, thereby enabling movement of the PPG sensor;

the silicone-based adhesive layer attaches to a first portion of the patient's skin; and

the acrylic-based adhesive layer attaches to a second portion of the patient's skin.

2. The patch of claim 1, wherein the patch comprises a vertical distance between the second side of the PPG sensor and the bottom surface area of the rigid support structure.

3. The patch of claim 2, wherein, when attached to the patient's skin, the second side of the PPG sensor is pressed into the patient's skin.

4. The patch of claim 3, wherein, when the patch is attached to the patient's skin, the patch on the patient, when the PPG sensor is pressed into the patient's skin, exerts an amount of pressure which is at least partially based on the vertical distance.

5. The patch of claim 4, wherein the vertical distance is 10 mm or less.

6. The patch of claim 1, wherein the compressible spacer is elastic.

7. The patch of claim 6, wherein the compressible spacer is silicone.

8. The patch of claim 7, wherein the compressible spacer is porous.

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9. The patch of claim 1, further comprising a flexible printed circuit board (PCB) connected to the first side of PPG sensor.

10. The patch of claim 9, wherein the PCB collects signal data from the PPG sensor.

11. The patch of claim 10, wherein the data from the PPG sensor is transmitted to a computing system.

12. The patch of claim 1, wherein the silicone-based adhesive layer defines a silicone-based adhesive layer opening and comprises a first side and a second side.

13. The patch of claim 12, wherein the acrylic-based adhesive layer is connected to the first side of the silicone-based adhesive layer, and wherein the acrylic-based adhesive layer is exposed on the second side of the silicone-based adhesive layer through the silicone-based adhesive layer opening.

14. The patch of claim 12, wherein the second portion of the patient's skin corresponds to at least a portion of the

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bottom surface area of the rigid support structure extending through the silicone-based adhesive layer opening.

15. The patch of claim 1, wherein the compressible spacer is a shape memory polymer configured to change the force on a patient's skin over time.

16. The patch of claim 1, wherein the enclosure is made of plastic.

17. The patch of claim 1, wherein the rigid support structure comprises a cradle and a pod.

18. The patch of claim 17, wherein the enclosure is formed by the cradle and/or the pod.

19. The patch of claim 18, wherein the rigid support structure, the silicone-based adhesive layer, and the acrylic-based adhesive layer form an area of strain isolation that includes the PPG sensor.

20. The patch of claim 1, wherein the PPG sensor is configured to remain in substantially constant contact with the patient's skin.

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